

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = total fields in the selected benchmark ADT (prefer parsed ADT definition; fall back to @BENCH adt_fields). SoA bufs = 1 tag buffer plus one buffer for each scalar field, plus one buffer for each non-self packed field annotated Linear, plus recursively counted buffers for each non-self packed field annotated Factored; self-recursive fields add no new buffers. Speedup >1× means the denominator is faster; **bold** marks >1.1×.

Program	ADT	SoA	End-to-end			Fold passes			Map passes		
	fields	bufs	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm
OctTree	16	9	0.6802	0.6650	1.02×	0.3735	0.3171	1.18 ×	0.3067	0.3480	0.88×
Compiler	9	8	0.2761	0.2116	1.30 ×	0.1130	0.0362	3.13 ×	0.1631	0.1754	0.93×
DBQuery	15	13	0.2635	0.2553	1.03×	0.1258	0.0877	1.43 ×	0.1377	0.1675	0.82×
DecisionTree	7	6	0.4801	0.2699	1.78 ×	0.4801	0.2699	1.78 ×	—	—	—
DomTree	15	14	0.2141	0.1519	1.41 ×	0.1545	0.0650	2.38 ×	0.0596	0.0869	0.69×
KDTree	17	16	0.1777	0.1395	1.27 ×	0.1777	0.1395	1.27 ×	—	—	—
LinearListReduction	11	11	0.0331	0.0050	6.59 ×	0.0331	0.0050	6.59 ×	—	—	—
reduceNestedList	3	3	0.0845	0.0073	11.53 ×	0.0845	0.0073	11.53 ×	—	—	—
List	2	2	0.5805	0.6434	0.90×	0.3258	0.3041	1.07×	0.2547	0.3393	0.75×
MonoTree	3	2	0.0620	0.0657	0.94×	0.0176	0.0222	0.79×	0.0444	0.0436	1.02×
ObjectGraph	5	4	0.2355	0.2404	0.98×	0.0932	0.0800	1.17 ×	0.1423	0.1605	0.89×
PiecewiseFunctions	8	7	0.3709	0.3512	1.06×	0.1345	0.0904	1.49 ×	0.2364	0.2609	0.91×
TernaryTree	5	3	0.1115	0.1050	1.06×	0.0288	0.0271	1.06×	0.0828	0.0779	1.06×
Trie	10	9	0.2173	0.2005	1.08×	0.0679	0.0327	2.08 ×	0.1494	0.1678	0.89×
ColorOctree	25	18	0.0886	0.0681	1.30 ×	0.0886	0.0681	1.30 ×	—	—	—

Table 2: Per-pass performance for OctTree, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers. Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med±err	SoA med±err	Speedup
barnesHutPotential	F	11/16	31%	0.0397±0.00027	0.0367±0.00230	1.08×
countActive	F	10/16	38%	0.0376±0.00189	0.0299±0.00001	1.26×
countParticles	F	8/16	50%	0.0360±0.00025	0.0264±0.00003	1.36×
fmmPotential	F	12/16	25%	0.0967±0.00004	0.0953±0.00210	1.01×
sumEnergy	F	12/16	25%	0.0383±0.00154	0.0336±0.00004	1.14×
sumMass	F	10/16	38%	0.0366±0.00252	0.0270±0.00001	1.35×
clearFlags	M	15/16	6%	0.1539±0.00228	0.1725±0.00167	0.89×
scaleEnergy	M	16/16	0%	0.1527±0.00036	0.1755±0.00204	0.87×
paletteEntriesQuantized	F	13/25	48%	0.0421±0.00189	0.0288±0.00002	1.46×
quantizationErrorProxy	F	10/25	60%	0.0465±0.00002	0.0393±0.00002	1.18×
Total				0.6802	0.6650	1.02×
Geomean						1.15×

Table 3: Per-pass performance for Compiler, ADT has 9 fields, SoA uses 8 buffers. Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med±err	SoA med±err	Speedup
blockCountPass	F	2/9	78%	0.0120±0.00001	0.0019±0.00000	6.23×
branchStatsPass	F	2/9	78%	0.0116±0.00001	0.0034±0.00000	3.41×
castInstCountPass	F	3/9	67%	0.0178±0.00001	0.0013±0.00000	13.83×
has cycle	F	4/9	56%	0.0242±0.00021	0.0178±0.00033	1.36×
instCountPass	F	2/9	78%	0.0120±0.00001	0.0019±0.00000	6.25×
latencyModelPass	F	3/9	67%	0.0119±0.00001	0.0030±0.00000	4.00×
memoryOpStatsPass	F	3/9	67%	0.0117±0.00001	0.0038±0.00001	3.07×
throughputModelPass	F	3/9	67%	0.0119±0.00001	0.0030±0.00000	3.96×
stripSideEffectsPass	M	7/9	22%	0.0844±0.00031	0.0865±0.00058	0.98×
targetReturnPass	M	9/9	0%	0.0787±0.00026	0.0890±0.00033	0.88×
Total				0.2761	0.2116	1.30×
Geomean						3.18×

Table 4: Per-pass performance for DBQuery, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med±err	SoA med±err	Speedup
countJoins	F	3/15	80%	0.0197±0.00001	0.0111±0.00001	1.78×
filterSelectivitySkew	F	5/15	67%	0.0195±0.00001	0.0125±0.00001	1.57×
hashJoinPressure	F	5/15	67%	0.0207±0.00001	0.0129±0.00002	1.60×
sumCost	F	6/15	60%	0.0197±0.00002	0.0146±0.00003	1.35×
sumMemory	F	6/15	60%	0.0195±0.00001	0.0148±0.00018	1.31×
sumRows	F	6/15	60%	0.0267±0.00002	0.0219±0.00003	1.22×
clearQueryFlags	M	14/15	7%	0.0682±0.00015	0.0831±0.00004	0.82×
scaleCosts	M	15/15	0%	0.0695±0.00027	0.0845±0.00218	0.82×
Total				0.2635	0.2553	1.03×
Geomean						1.26×

Table 5: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers. Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med±err	SoA med±err	Speedup
classify Batch	F	5/7	29%	0.00±0.000	0.00±0.000	0.72×
countFeatureUses	F	3/7	57%	0.0538±0.00003	0.0326±0.00003	1.65×
countLeaves	F	2/7	71%	0.0505±0.00003	0.0247±0.00002	2.04×
countNodes	F	2/7	71%	0.0510±0.00002	0.0238±0.00002	2.15×
countSmallLeaves	F	3/7	57%	0.0548±0.00005	0.0322±0.00006	1.70×
inferenceCost	F	2/7	71%	0.0539±0.00002	0.0287±0.00001	1.88×
sumImpurity	F	3/7	57%	0.0538±0.00003	0.0328±0.00003	1.64×
sumPathLengths	F	3/7	57%	0.0561±0.00004	0.0340±0.00003	1.65×
sumSamples	F	3/7	57%	0.0514±0.00003	0.0324±0.00003	1.59×
treeDepth	F	2/7	71%	0.0547±0.00037	0.0287±0.00002	1.91×
Total				0.4801	0.2699	1.78×
Geomean						1.64×

Table 6: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
count styled	F	3/15	80%	0.0383 $\pm$ 0.00002	0.0116$\pm$0.00001	3.31$\times$
find max Bottom	F	5/15	67%	0.0385 $\pm$ 0.00108	0.0226$\pm$0.00004	1.70$\times$
SumArea	F	6/15	60%	0.0398 $\pm$ 0.00004	0.0191$\pm$0.00157	2.08$\times$
sumTextWidth	F	3/15	80%	0.0380 $\pm$ 0.00002	0.0117$\pm$0.00001	3.25$\times$
computeWidths	M	14/15	7%	0.0307$\pm$0.00012	0.0472 $\pm$ 0.00013	0.65 $\times$
scaleLayout	M	15/15	0%	0.0289$\pm$0.00005	0.0398 $\pm$ 0.00008	0.73 $\times$
Total				0.2141	0.1519	1.41$\times$
Geomean						1.62$\times$

Table 7: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countInRange tight_box	F	13/17	24%	0.0276 $\pm$ 0.00002	0.0199$\pm$0.00002	1.39$\times$
Find nearest Neighbour	F	13/17	24%	0.0275 $\pm$ 0.00003	0.0213$\pm$0.00002	1.29$\times$
photonMappingBenchmark	F	12/17	29%	0.0423 $\pm$ 0.00002	0.0412 $\pm$ 0.00003	1.03 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0249 $\pm$ 0.00001	0.0166$\pm$0.00001	1.50$\times$
sumMassInRange	F	14/17	18%	0.0271 $\pm$ 0.00002	0.0200$\pm$0.00165	1.36$\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0284 $\pm$ 0.00001	0.0206$\pm$0.00001	1.38$\times$
Total				0.1777	0.1395	1.27$\times$
Geomean						1.31$\times$

Table 8: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/11	82%	0.0331 $\pm$ 0.00001	0.0050$\pm$0.00001	6.59$\times$
Total				0.0331	0.0050	6.59$\times$
Geomean						6.59$\times$

Table 9: Per-pass performance for **reduceNestedList**, ADT has 3 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/3	33%	0.0845 $\pm$ 0.00008	0.0073$\pm$0.00001	11.53$\times$
Total				0.0845	0.0073	11.53$\times$
Geomean						11.53$\times$

Table 10: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
length List	F	1/2	50%	0.0494 $\pm$ 0.00009	0.0264$\pm$0.00002	1.87$\times$
sumList	F	2/2	0%	0.1382 $\pm$ 0.00004	0.1387 $\pm$ 0.00326	1.00 $\times$
sumList tail recursive	F	2/2	0%	0.1383 $\pm$ 0.00004	0.1390 $\pm$ 0.00025	0.99 $\times$
add1 List	M	2/2	0%	0.2547$\pm$0.00270	0.3393 $\pm$ 0.00222	0.75 $\times$
Total				0.5805	0.6434	0.90 $\times$
Geomean						1.09 $\times$

Table 11: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sumTree	F	3/3	0%	0.0090$\pm$0.00001	0.0113 $\pm$ 0.00002	0.80 $\times$
sumTree TailRec	F	3/3	0%	0.0086$\pm$0.00001	0.0108 $\pm$ 0.00004	0.79 $\times$
add1Tree	M	3/3	0%	0.0444 $\pm$ 0.00208	0.0436 $\pm$ 0.00012	1.02 $\times$
Total				0.0620	0.0657	0.94 $\times$
Geomean						0.86 $\times$

Table 12: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countLargeItems	F	3/5	40%	0.0130 $\pm$ 0.00001	0.0103$\pm$0.00003	1.25$\times$
countMarked	F	3/5	40%	0.0129 $\pm$ 0.00001	0.0105$\pm$0.00003	1.22$\times$
countSurvivors	F	4/5	20%	0.0139 $\pm$ 0.00001	0.0132 $\pm$ 0.00001	1.06 $\times$
deadBytes	F	4/5	20%	0.0143 $\pm$ 0.00001	0.0126$\pm$0.00001	1.14$\times$
liveBytes	F	4/5	20%	0.0135 $\pm$ 0.00001	0.0126 $\pm$ 0.00001	1.07 $\times$
sumObjIds	F	3/5	40%	0.0129 $\pm$ 0.00001	0.0102$\pm$0.00003	1.26$\times$
totalHeapSize	F	3/5	40%	0.0127 $\pm$ 0.00001	0.0104$\pm$0.00003	1.22$\times$
sweepUnmarked	M	5/5	0%	0.0715$\pm$0.00025	0.0806 $\pm$ 0.00167	0.89 $\times$
touchHotObjects	M	5/5	0%	0.0708$\pm$0.00199	0.0799 $\pm$ 0.00004	0.89 $\times$
Total				0.2355	0.2404	0.98 $\times$
Geomean						1.10$\times$

Table 13: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autorefineMaxLevel	F	4/8	50%	0.0209 $\pm$ 0.00002	0.0187$\pm$0.00009	1.12$\times$
compressMass	F	3/8	62%	0.0187 $\pm$ 0.00002	0.0116$\pm$0.00002	1.61$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0206 $\pm$ 0.00001	0.0177$\pm$0.00003	1.17$\times$
norm2Estimate	F	5/8	38%	0.0242 $\pm$ 0.00003	0.0182$\pm$0.00003	1.33$\times$
pmapCutHistogram	F	4/8	50%	0.0312 $\pm$ 0.00032	0.0127$\pm$0.00123	2.46$\times$
truncateTolViolations	F	3/8	62%	0.0190 $\pm$ 0.00002	0.0114$\pm$0.00002	1.67$\times$
addConstPW	M	8/8	0%	0.1186$\pm$0.00053	0.1319 $\pm$ 0.00048	0.90 $\times$
diffPW	M	8/8	0%	0.1178 $\pm$ 0.00232	0.1289 $\pm$ 0.00017	0.91 $\times$
Total				0.3709	0.3512	1.06 $\times$
Geomean						1.32$\times$

Table 14: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sum tree	F	5/5	0%	0.0288 $\pm$ 0.00012	0.0271 $\pm$ 0.00010	1.06 $\times$
add 1 tree	M	5/5	0%	0.0828 $\pm$ 0.00279	0.0779 $\pm$ 0.00014	1.06 $\times$
Total				0.1115	0.1050	1.06 $\times$
Geomean						1.06 $\times$

Table 15: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autocompleteTopKProxy	F	5/10	50%	0.0154 $\pm$ 0.00001	0.0087$\pm$0.00003	1.77$\times$
countLazyNodes	F	4/10	60%	0.0136 $\pm$ 0.00002	0.0067$\pm$0.00001	2.04$\times$
countTerminals	F	3/10	70%	0.0128 $\pm$ 0.00001	0.0058$\pm$0.00001	2.21$\times$
sumPrefixFreq	F	3/10	70%	0.0130 $\pm$ 0.00001	0.0058$\pm$0.00001	2.25$\times$
sumSubtreeHints	F	3/10	70%	0.0130 $\pm$ 0.00001	0.0057$\pm$0.00001	2.27$\times$
decayTrieStats	M	10/10	0%	0.0750$\pm$0.00025	0.0859 $\pm$ 0.00231	0.87 $\times$
resetTraversalState	M	10/10	0%	0.0745 $\pm$ 0.00191	0.0819 $\pm$ 0.00261	0.91 $\times$
Total				0.2173	0.2005	1.08 $\times$
Geomean						1.64$\times$